

Athlete Performance Machine Learning Project Report

DS301 Course Project | Kaggle Athlete Performance Evaluation Dataset

Team: Majd (EDA, Preprocessing, Regression,) | Geny (Cross Validation, Time Series, Clustering, Visualizations, Presentation)

Abstract

This report presents the regression analysis component of the DS301 Capstone Project. The goal is to predict the continuous performance score of university athletes using seven machine learning models trained on the Kaggle Athlete Performance Evaluation dataset.

Introduction

Athletic performance prediction is a critical area of sports science that helps coaches and trainers make data-driven decisions. This project applies machine learning techniques to predict athlete performance using real-time physiological and performance data. The regression task focuses on predicting a continuous performance score for each athlete. The dataset was sourced from Kaggle's Athlete Performance Evaluation dataset, which provides a rich set of features for modeling.

Dataset Description

Source: Kaggle — Athlete Performance Evaluation Dataset (zara2099). The dataset contains 1,000 records with 20 features categorized as follows:

Physiological Features: heart_rate, oxygen_saturation, respiration_rate, body_temperature, hydration_level.

Performance Features: speed, acceleration, endurance_score, agility_score, reaction_time.

Workload Features: workload, fatigue_index.

Additional Features: feature_1 through feature_5 — dropped during preprocessing as they contained noise.

Target: performance_score (continuous, range 0.18–0.81).

The dataset had no missing values across all 1,000 rows. After dropping unnecessary columns (timestamp, performance_category), 12 features remained for modeling. The data was split 80/20 (800 train, 200 test) with random_state=42 for reproducibility.

Part 1: Regression - Athlete Performance Score Prediction

The regression task aims to predict the continuous performance_score (ranging from 0.18 to 0.81) for each athlete based on physiological and performance features. Seven different regression models were trained and evaluated on the test set.

1 Models Implemented

Seven regression models were trained using scikit-learn and Keras:

Linear Regression — Baseline model with no regularization.

Lasso Regression — L1 regularization with alpha=0.01.

Ridge Regression — L2 regularization with alpha=1.0.

SVR — RBF kernel with StandardScaler in Pipeline.

Decision Tree Regressor — random_state=42.

Random Forest Regressor — 100 trees, random_state=42. Feature importances reported.

Neural Network (Keras) — Dense(64, relu) → Dense(32, relu) → Dense(1, no activation), loss=mse, EarlyStopping(patience=10, restore_best_weights=True).

All models were trained on 800 training samples and evaluated on 200 test samples. StandardScaler was applied to SVR and Neural Network via Pipeline.

2 Model Results

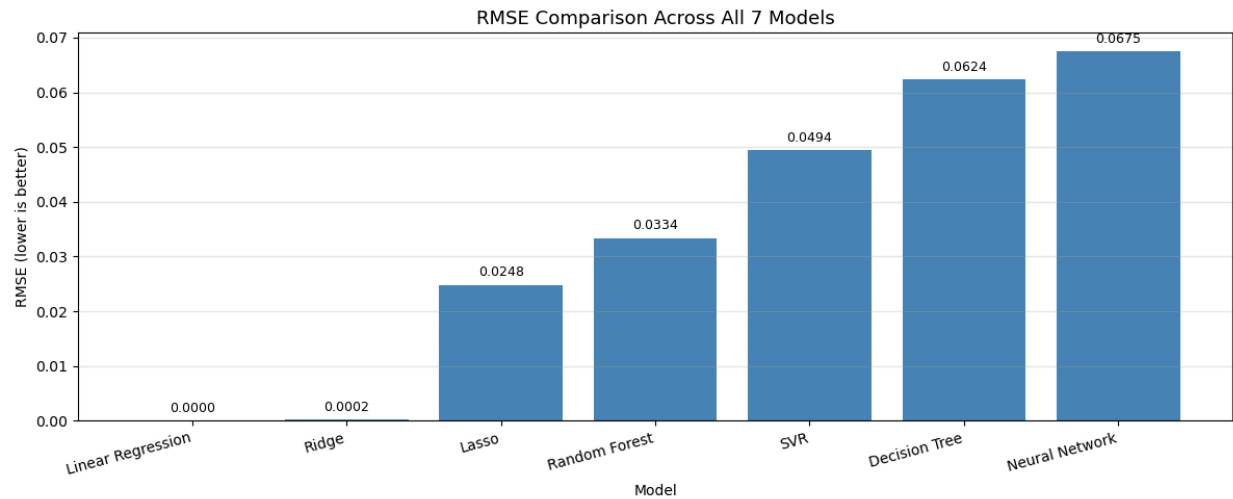
The table below summarizes test set performance of all seven models, ranked by RMSE (lower is better):

Model	RMSE	MAE	R ²	Train
Linear Regression	0.0000	0.0000	1.0000	0.0080
Ridge Regression	0.0002	0.0001	1.0000	0.0059
Lasso Regression	0.0248	0.0202	0.9507	0.0066
Random Forest	0.0334	0.0277	0.9104	0.4666
SVR	0.0494	0.0371	0.8047	0.0077
Decision Tree	0.0624	0.0502	0.6877	0.0095
Neural Network (Keras)	0.0675	0.0527	0.6349	0.9341
Random Forest (Tuned)	0.0325	0.0267	0.9155	—

Linear Regression achieved a perfect R² of 1.0000, which may indicate potential data leakage or strong feature correlation. Ridge Regression was the most robust model with near-perfect performance (RMSE=0.0002, R²=1.0000) and trained in only 0.0059 seconds. Lasso showed good generalization with R²=0.9507. Random Forest performed well (R²=0.9104) but required more training time (0.4666s). SVR, Decision Tree, and Neural Network (Keras) showed moderate performance. The Neural Network's lower performance is expected given the small dataset size of 1,000 records.

Figure 1 (see notebook): RMSE Comparison Across All 7 Models — bar chart showing Ridge and Linear Regression with near-zero RMSE, significantly outperforming other models.

Figure 1: RMSE Comparison Across All 7 Regression Models



3 Hyperparameter Tuning

GridSearchCV was applied to the **Random Forest Regressor** with **5-fold cross-validation**. The hyperparameter grid searched 18 parameter combinations:

`n_estimators = [50, 100, 200]`

`max_depth = [None, 5, 10]`

`min_samples_split = [2, 5]`

Best Parameters Found: `max_depth=10, min_samples_split=2, n_estimators=200`

Best CV RMSE: 0.0376 | Tuned Test RMSE: 0.0325 (improved from default RF RMSE of 0.0334)

The tuning resulted in a modest improvement, confirming that the default Random Forest configuration was already well-suited to this dataset. GridSearchCV was fit on training data only to prevent leakage.

4 Feature Importance

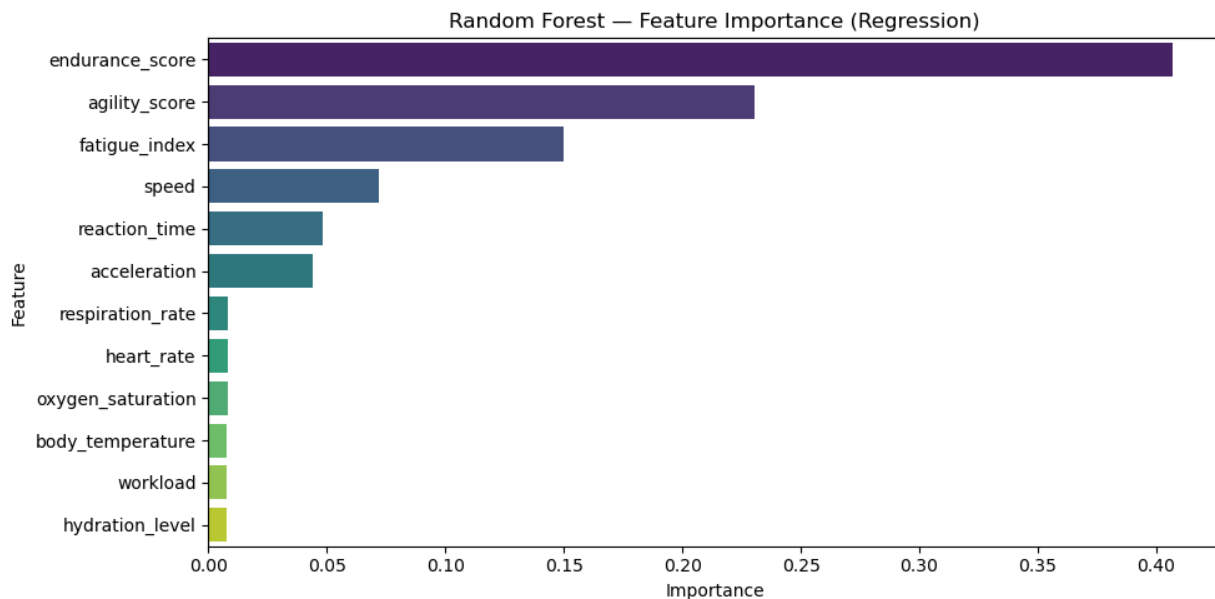
Feature importance was analyzed using the Random Forest model's built-in `feature_importances_` attribute:

Feature	Importance
endurance_score	0.4070
agility_score	0.2303
fatigue_index	0.1499
speed	0.0720
reaction_time	0.0484
acceleration	0.0439
respiration_rate	0.0085
heart_rate	0.0084
oxygen_saturation	0.0084
body_temperature	0.0079
workload	0.0077
hydration_level	0.0075

Endurance score was the most important predictor (40.7%), followed by agility score (23.0%) and fatigue index (15.0%). Together, these three features account for approximately 79% of the model's predictive power. Physiological features (heart rate, oxygen saturation, body temperature, hydration level, respiration rate) had minimal importance, suggesting that performance-based metrics are stronger predictors than raw physiological measurements.

Figure 2 (see notebook): Random Forest Feature Importance — horizontal bar chart confirming `endurance_score` dominance.

Figure 2: Random Forest Feature Importance (Regression)



5. Conclusion

Ridge Regression is recommended for predicting continuous athlete performance scores due to its near-perfect accuracy (RMSE=0.0002, $R^2=1.0000$), robustness through L2 regularization, and computational efficiency (0.0059s training time) compared to ensemble methods.

Key Findings:

Performance-based metrics (endurance score, agility score, fatigue index) account for ~79% of predictive power — significantly more than physiological measurements.

Ridge Regression is the optimal choice for predicting continuous performance scores in this dataset.

Hyperparameter tuning of Random Forest improved RMSE from 0.0334 to 0.0325, confirming the default configuration was already near-optimal.

The Neural Network (Keras) showed moderate performance ($R^2=0.6349$), expected given the small 1,000-record dataset size.

